

EE4C10

Analog Circuit Design

Fundamentals

Week 2

Qinwen Fan

Substitute lecturer today: Mengying

Today's goal

- Recap basic concepts and the fundamentals of CS stage with quizzes
- Go through 2 LT spice exercises

Acknowledgments

Some images from B.Razavi's "Design of analog CMOS integrated circuits", copyright of McGraw-Hill.

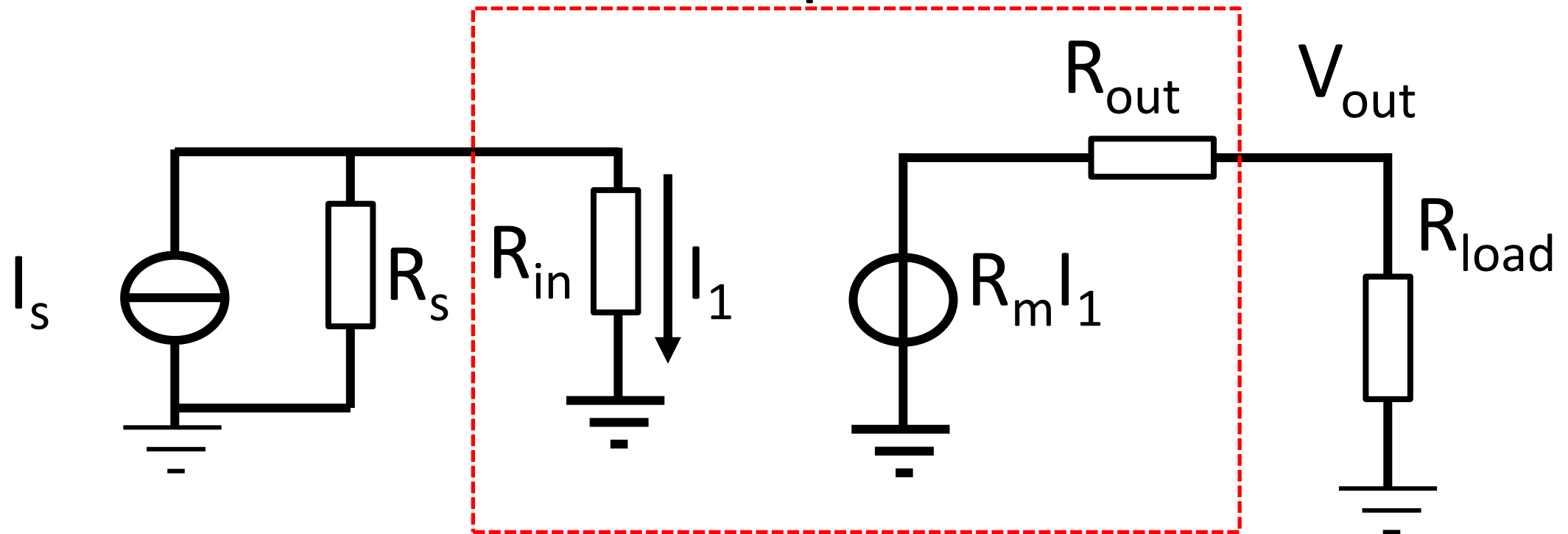
Q1

- What is the desired input and output impedance of a current-voltage amplifier
 - A. High input impedance; high output impedance
 - B. High input impedance; low output impedance
 - C. Low input impedance; high output impedance
 - D. Low input impedance; low output impedance

Q1

- What is the desired input and output impedance of a current-voltage amplifier
 - A. High input impedance; high output impedance
 - B. High input impedance; low output impedance
 - C. Low input impedance; high output impedance
 - D. Low input impedance; low output impedance

Q1 Amplifier



Above is a general schematic of a current-voltage amplifier.

- In order to “absorb” all the signal current I_s , R_{in} should be ideally 0.
- In order to apply $R_m I_1$ (the amplified voltage signal) to the load, R_{out} should be ideally 0

Q2

- What kind of simulation is usually used to check the bias condition of an amplifier
 - A. AC simulation
 - B. DC simulation
 - C. Transient simulation
 - D. None of above

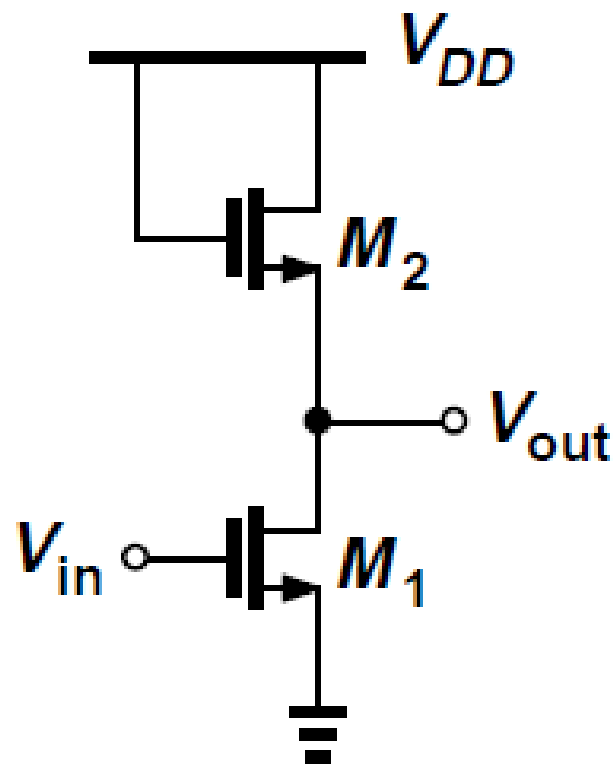
Q2

- What kind of simulation is usually used to check the bias condition of an amplifier
 - A. AC simulation
 - B. DC simulation**
 - C. Transient simulation
 - D. None of above

Always check the bias condition of your amplifier before you do AC and transient simulations. Make sure your transistors are properly biased by checking their V_{GS} , V_{DS} , drain current, transconductance, and etc.

Q3

- What is the output voltage swing of this amplifier, to keep both transistors in saturation?
 - A. $V_{dsat1} \sim (V_{DD} - V_{th2})$
 - B. $0 \sim V_{dsat2}$
 - C. $V_{dsat1} \sim (V_{DD} - V_{dsat2})$
 - D. $0 \sim V_{DD}$



Q3

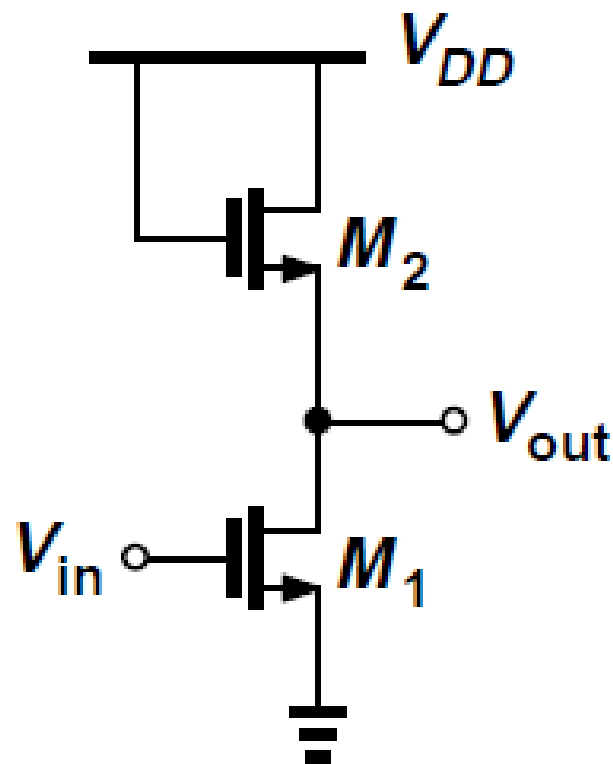
- What is the output voltage swing of this amplifier, to keep both transistors in saturation?

A. $V_{dsat1} \sim (V_{DD} - V_{th2})$

B. $0 \sim V_{dsat2}$

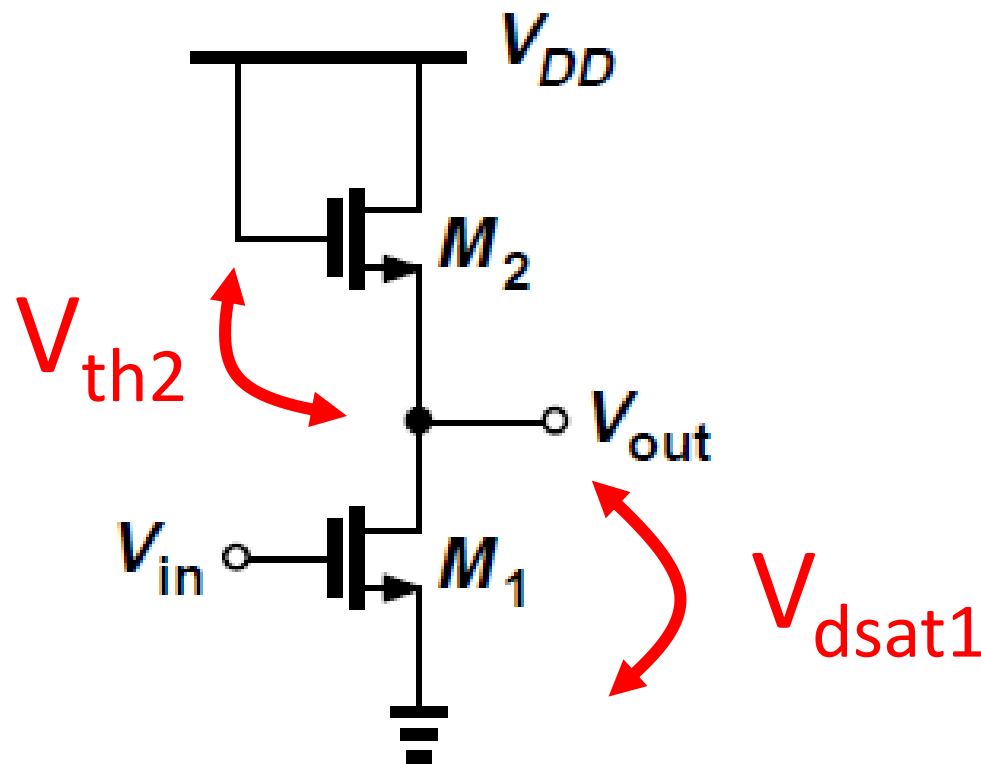
C. $V_{dsat1} \sim (V_{DD} - V_{dsat2})$

D. $0 \sim V_{DD}$



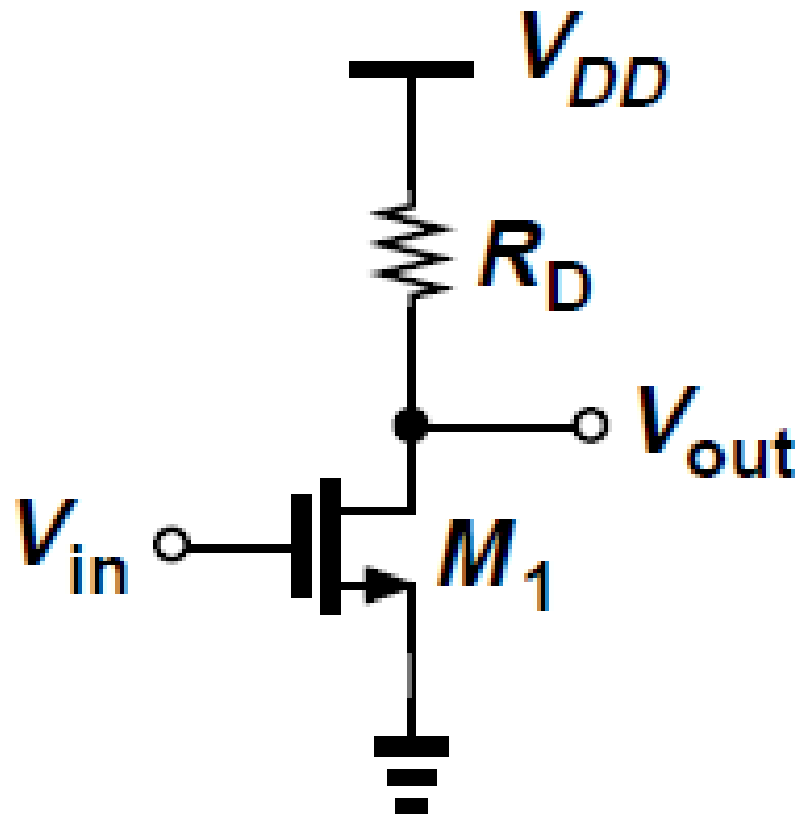
Q3

- What is the output voltage swing of this amplifier, to keep both transistors in saturation?
- To keep M_1 in saturation, V_{out} should be higher than V_{dsat1} .
- $V_{dsat} = V_{gs} - V_{th}$
- To keep M_2 on, V_{out} should be lower than $V_{DD} - V_{th2}$



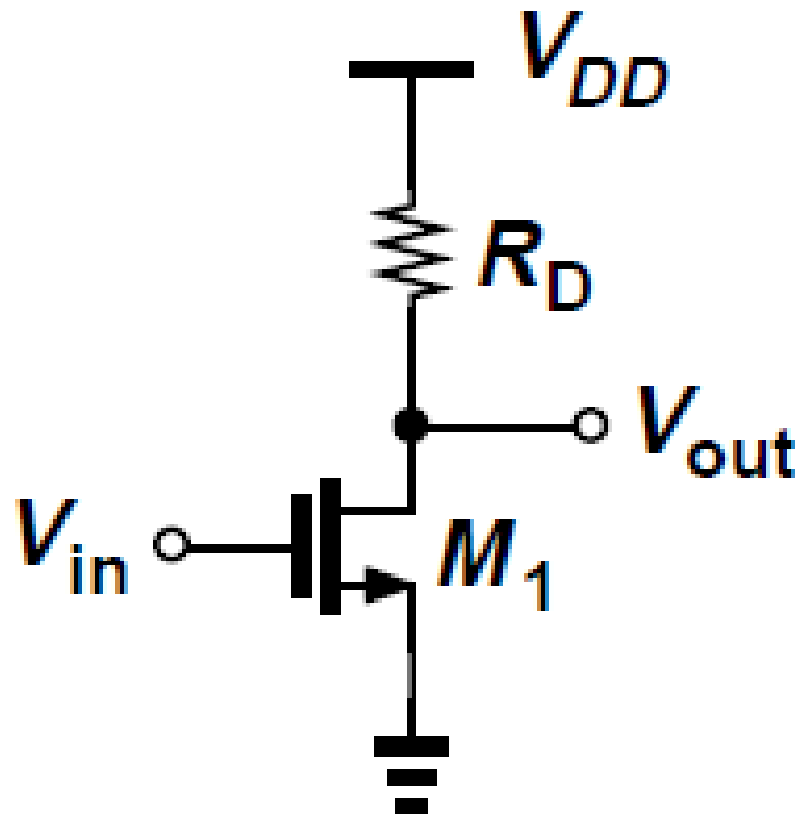
Q4

- What would you do to increase the gain of this amplifier, assume R_D is much smaller than r_{o1} ?
 - A. Replace R_D by a current source
 - B. Decrease the transconductance of M_1
 - C. None of above



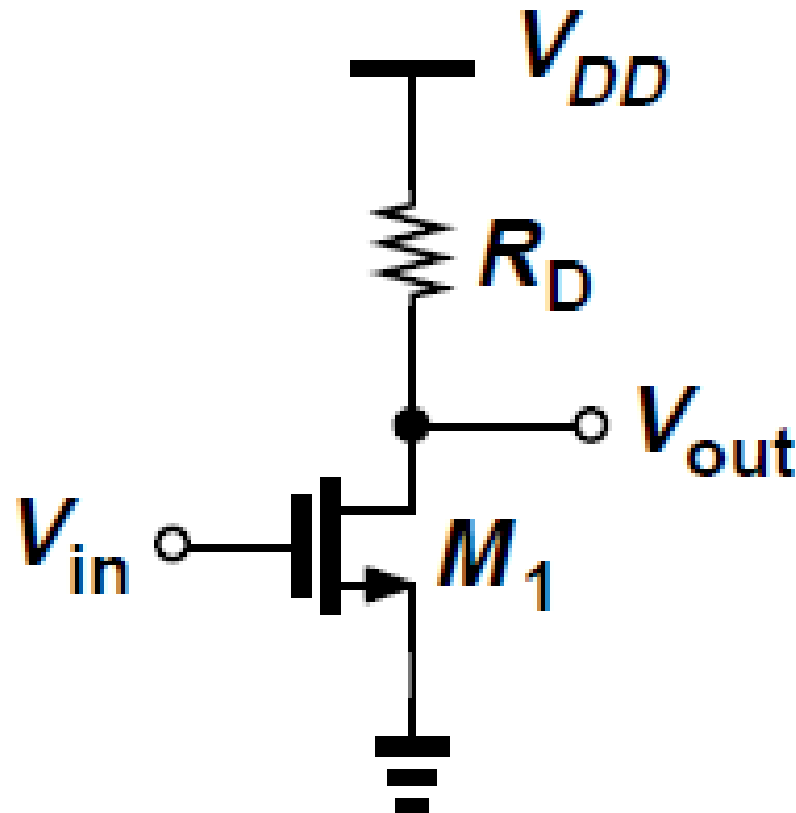
Q4

- What would you do to increase the gain of this amplifier, assume R_D is much smaller than r_{o1} ?
 - A. Replace R_D by a current source
 - B. Decrease the transconductance of M_1
 - C. None of above



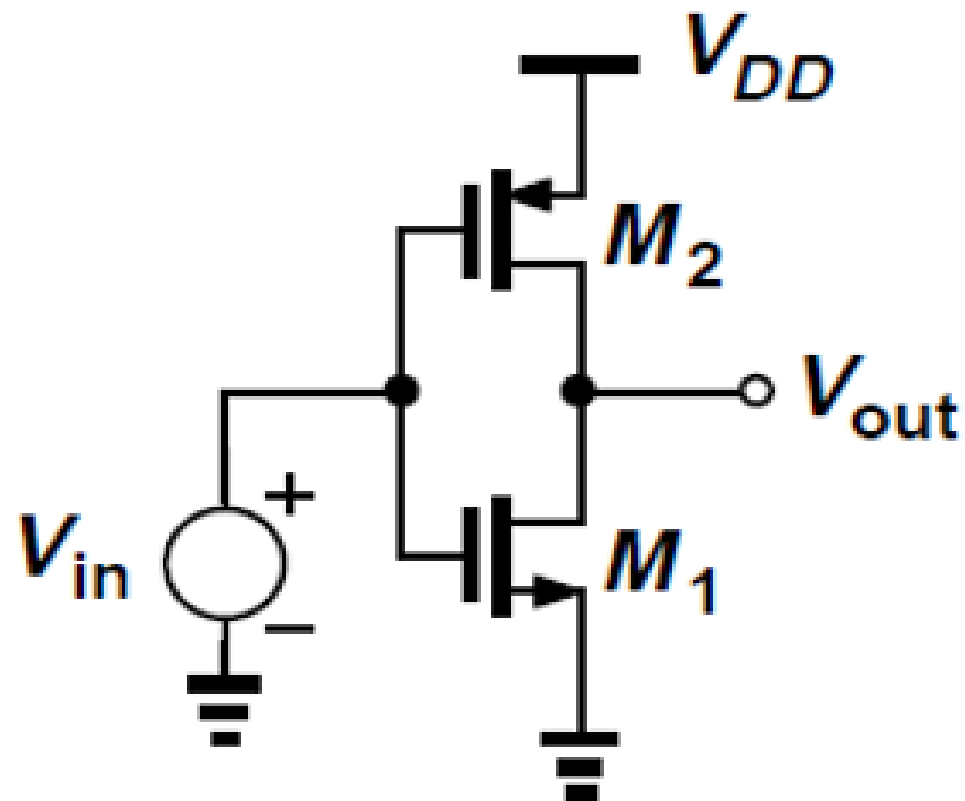
Q4

- The gain of the amplifier is $\sim g_m R_D$, since r_{o1} is much larger than R_D
- If we replace R_D with a current source, we assume the output impedance of that current source r_o can be much larger than R_D , therefore increase gain.



Q5

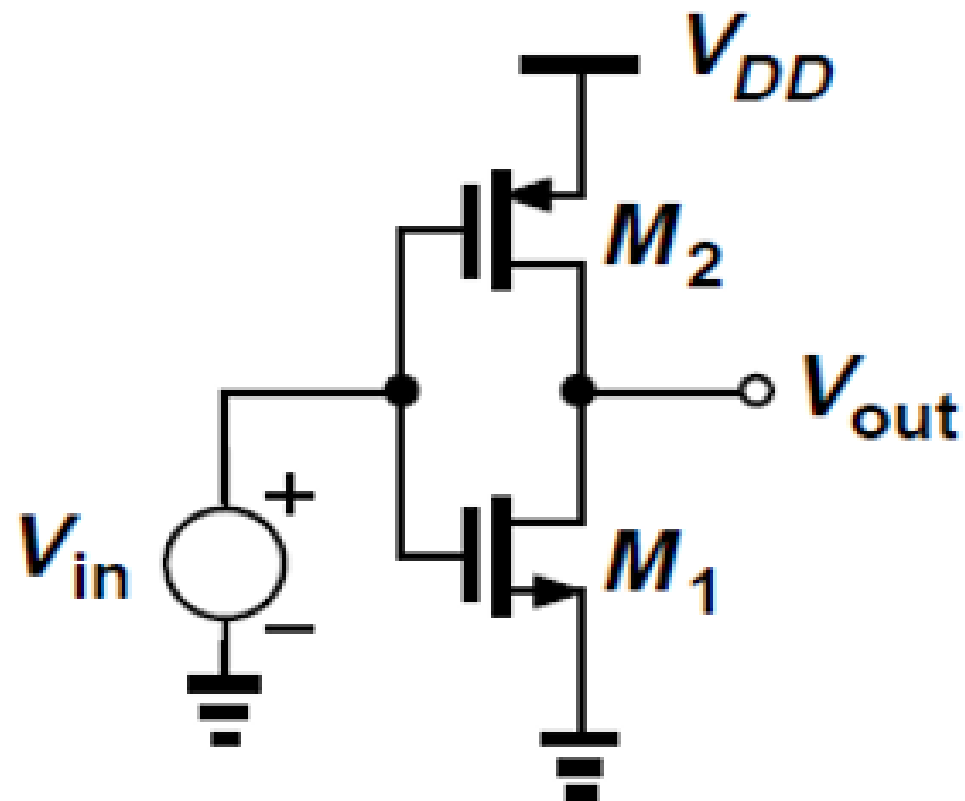
- What are the advantages of inverter-based amplifier?
 - A. High gain
 - B. Low supply sensitivity
 - C. Low PVT variation



Q5

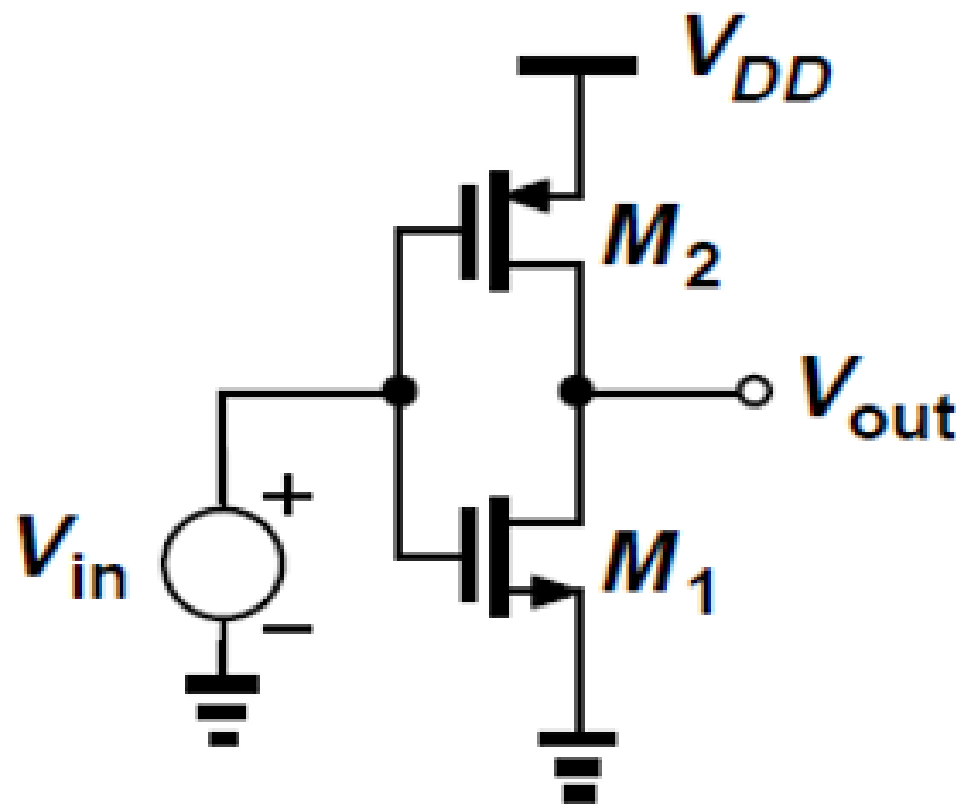
- What are the advantages of inverter-based amplifier?

- A. High gain
- B. Low supply sensitivity
- C. Low PVT variation



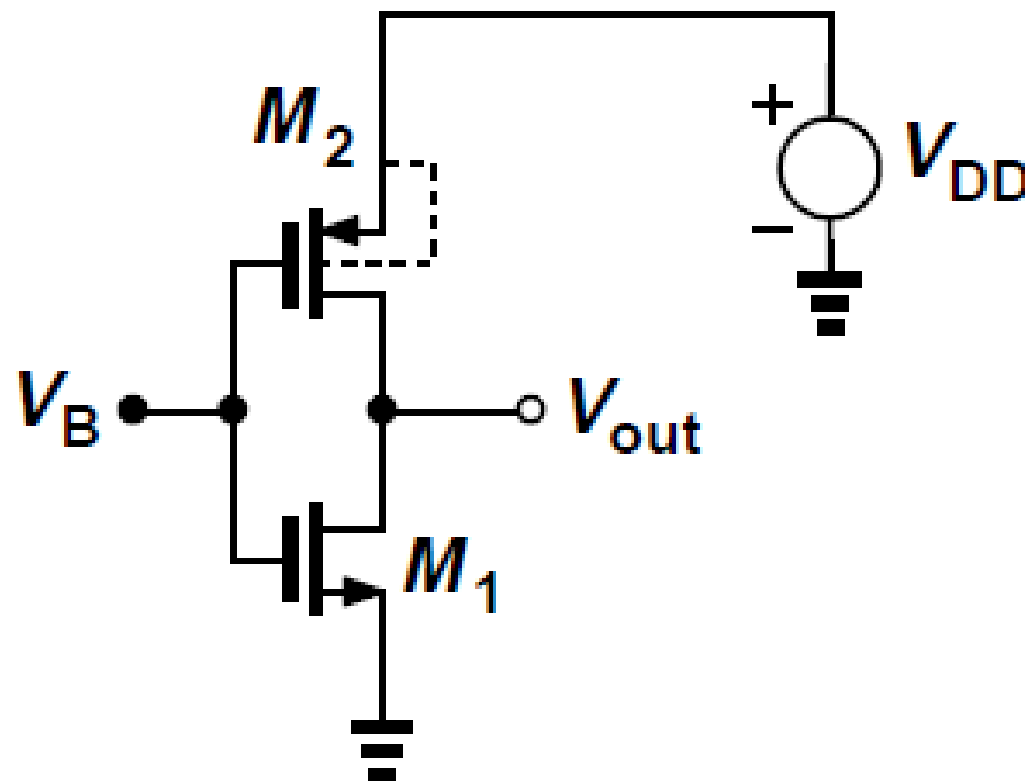
Q5

- The gain of the inverter-based amplifier is $-(g_{m1}+g_{m2}) \cdot r_{o1} // r_{o2}$. compared to a normal CS stage with a current source load, it offers a higher total transconductance while the output impedance is similar.



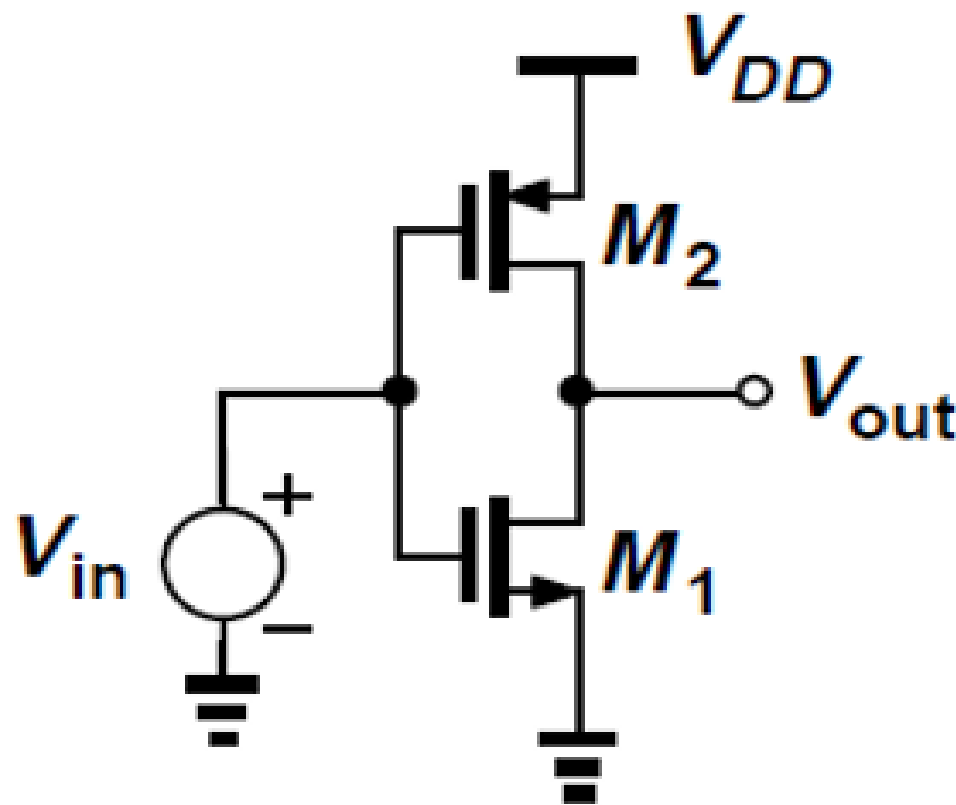
Q5

- Inverter-based amplifiers, however, are very sensitive to supply noise. Any supply noise will be amplified by M_2 . So in the reality, we need a clean V_{DD} when we use inverter-based amplifiers.



Q5

- The gain of the inverter-based amplifier is quite PVT dependent, because g_{m1} , g_{m2} and r_o all depends on PVT.



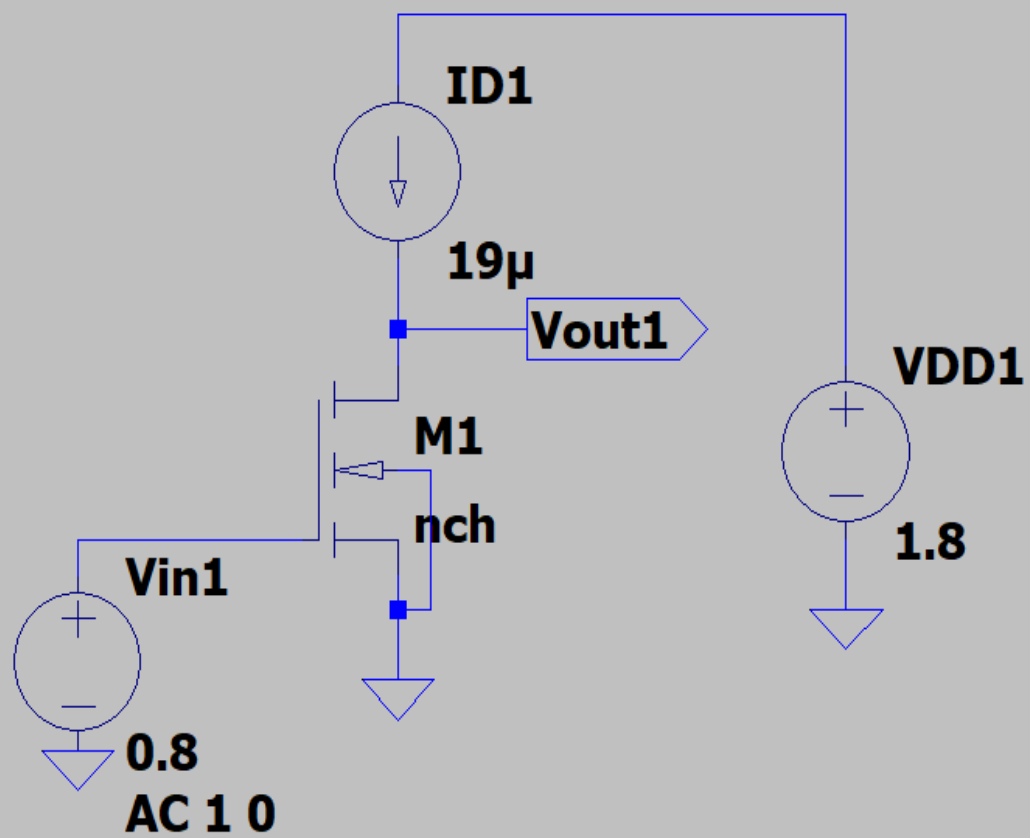
LTSpice Simulation Session

- We have 2 exercises.
- Download the exercises from bright space:
- LTspice files for practicals=> Week 2=> Week2 Exc1, Week2 Exc2

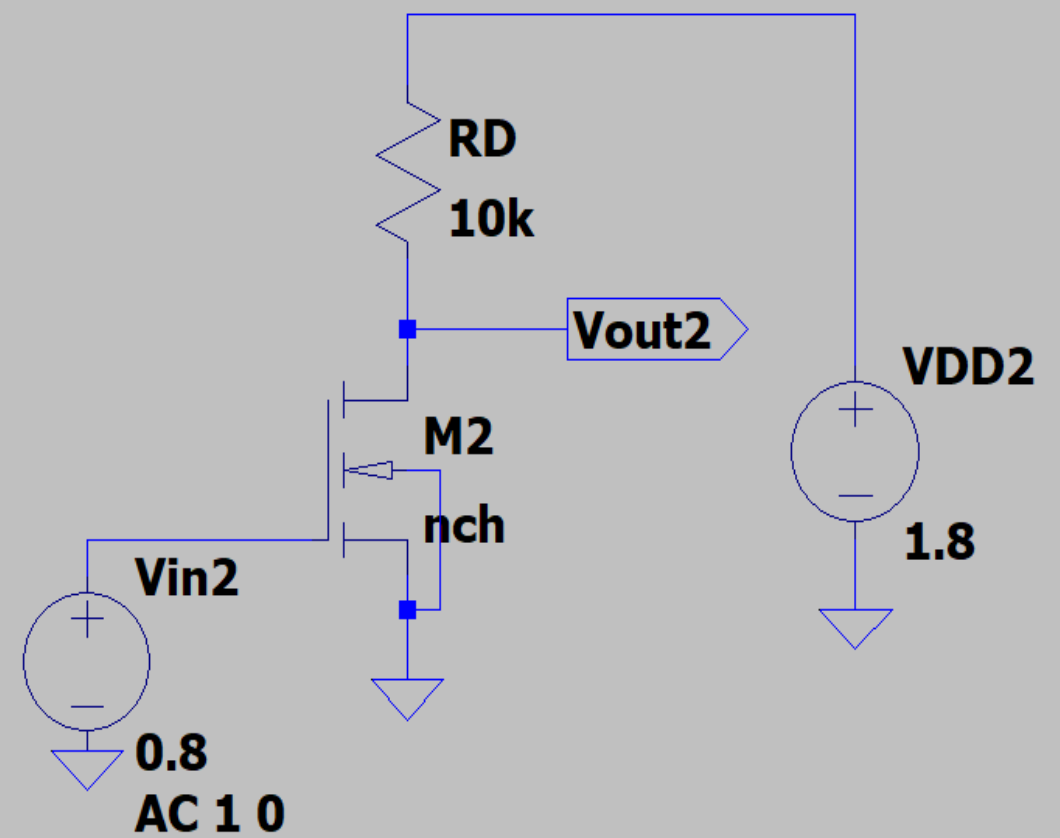
Exercise 1a

- Open Exc1.asc, you will see two CS stage amplifiers. Amplifier A has a current source load and amplifier B has a resistor load.
- Do a DC operating point simulation to find out whether M_1 and M_2 are biased in saturation.
- Calculate the DC gain of both amplifiers.
- Do an AC simulation to plot the gain of the two amplifiers. Does the simulation agree with your calculation?

Exercise 1a



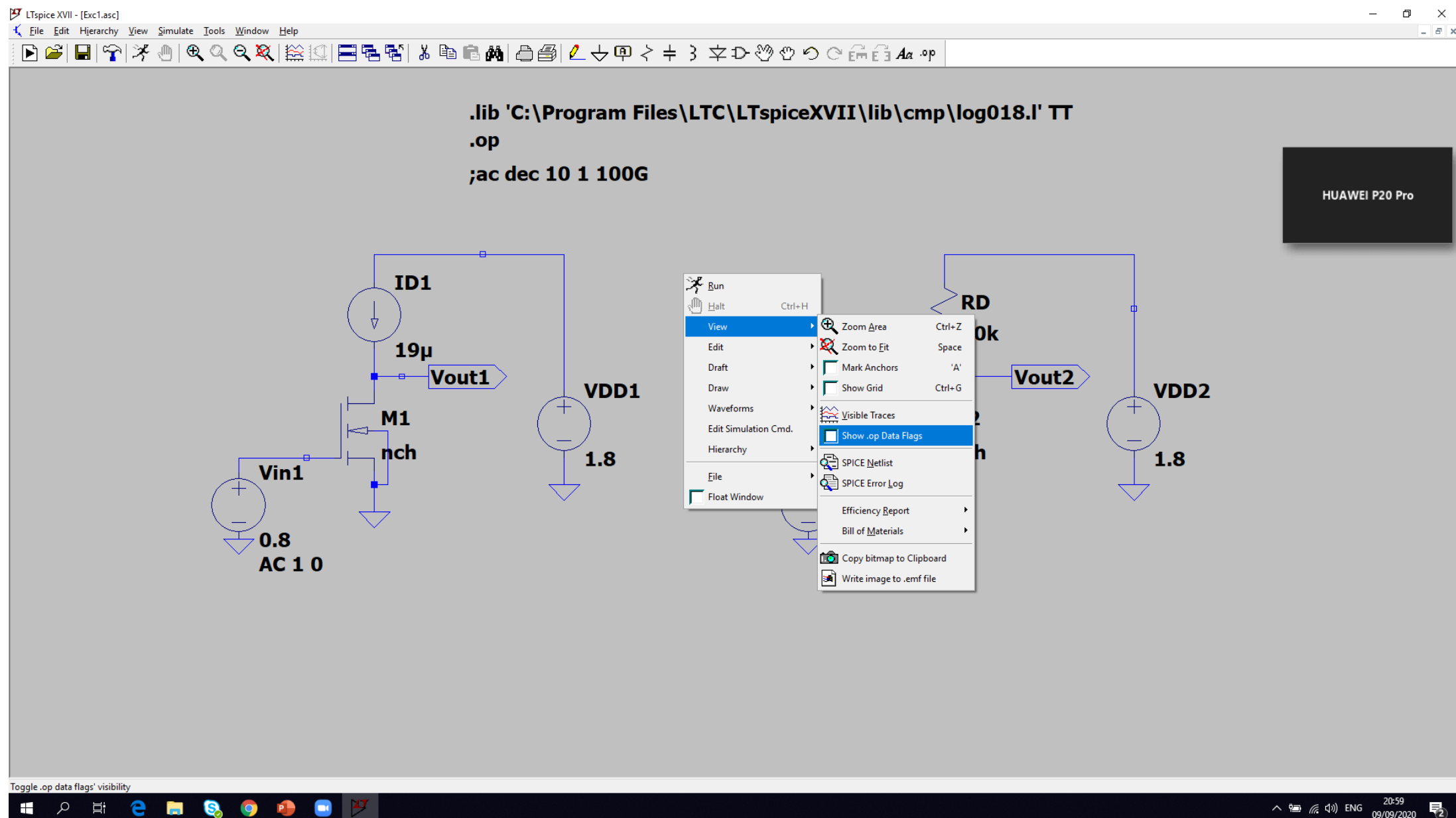
A



B

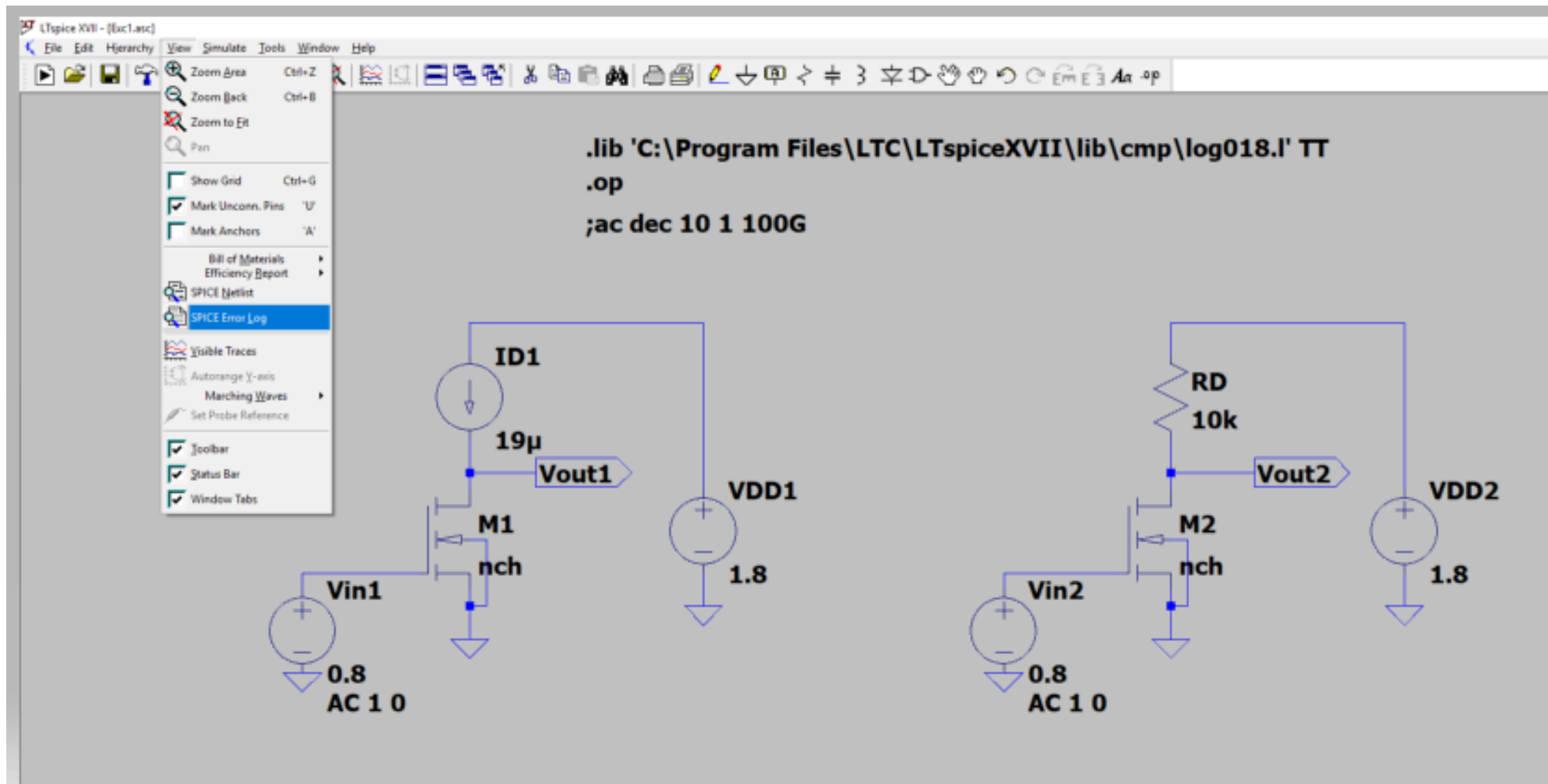
Exercise 1a

Tips: To annotate the DC voltages on the schematic, right click on the schematic => view => show .op Data Flags => left click on the nodes you want to check



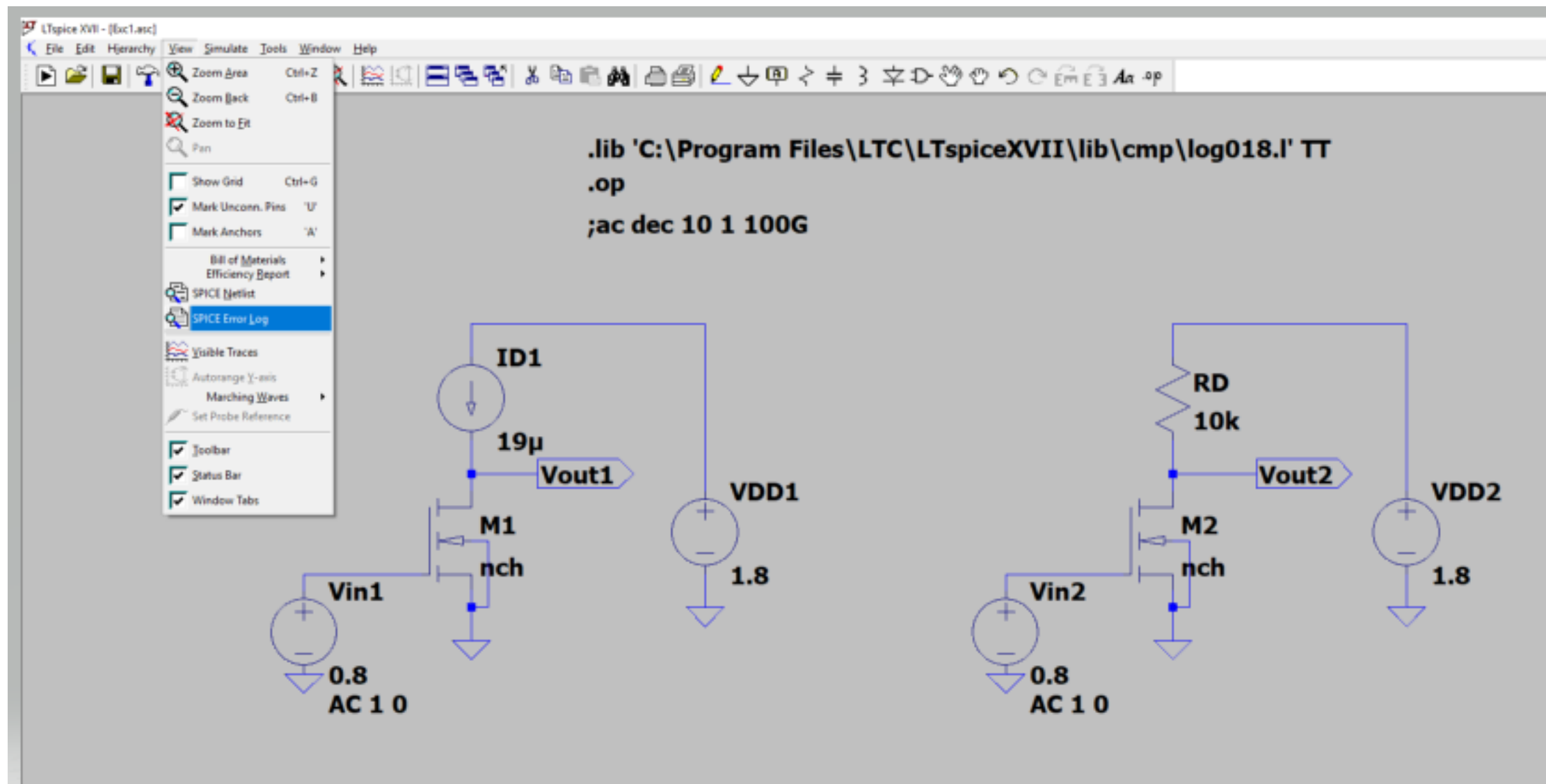
Exercise 1a

Tips: To print the DC operating point of the entire circuit: run .op simulation => Go to the tool bar “View” => SPICE error log



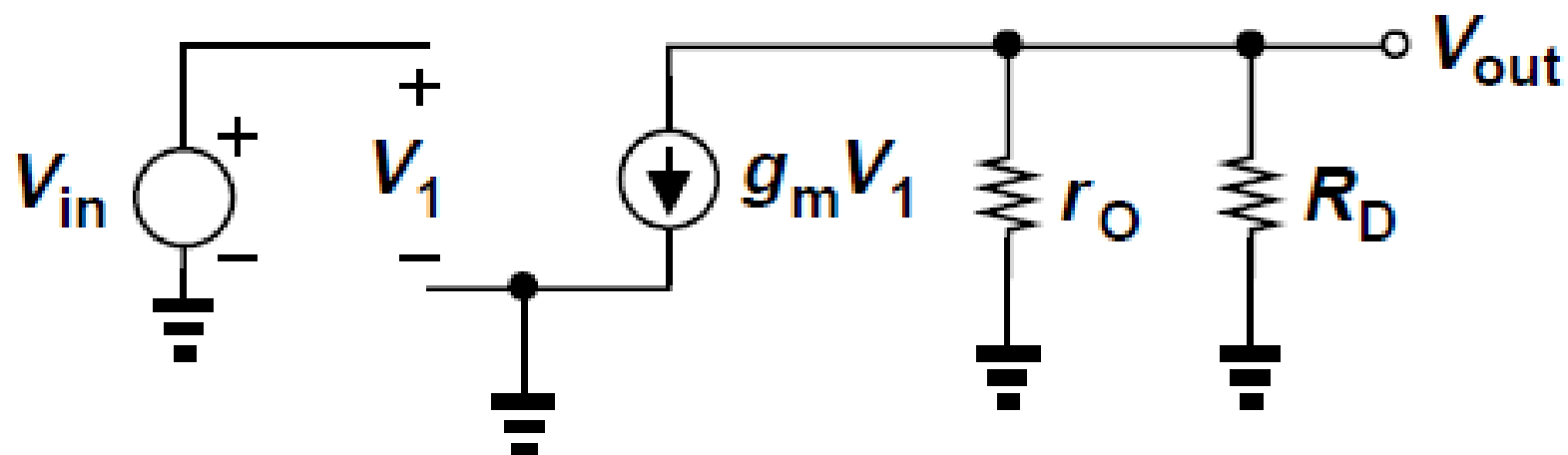
Exercise 1a

Tips: To calculate the DC gain, you need to find out g_m and r_o . $r_o = 1/G_{ds}$



Theoretical Material: CS Stage Gain Calculation

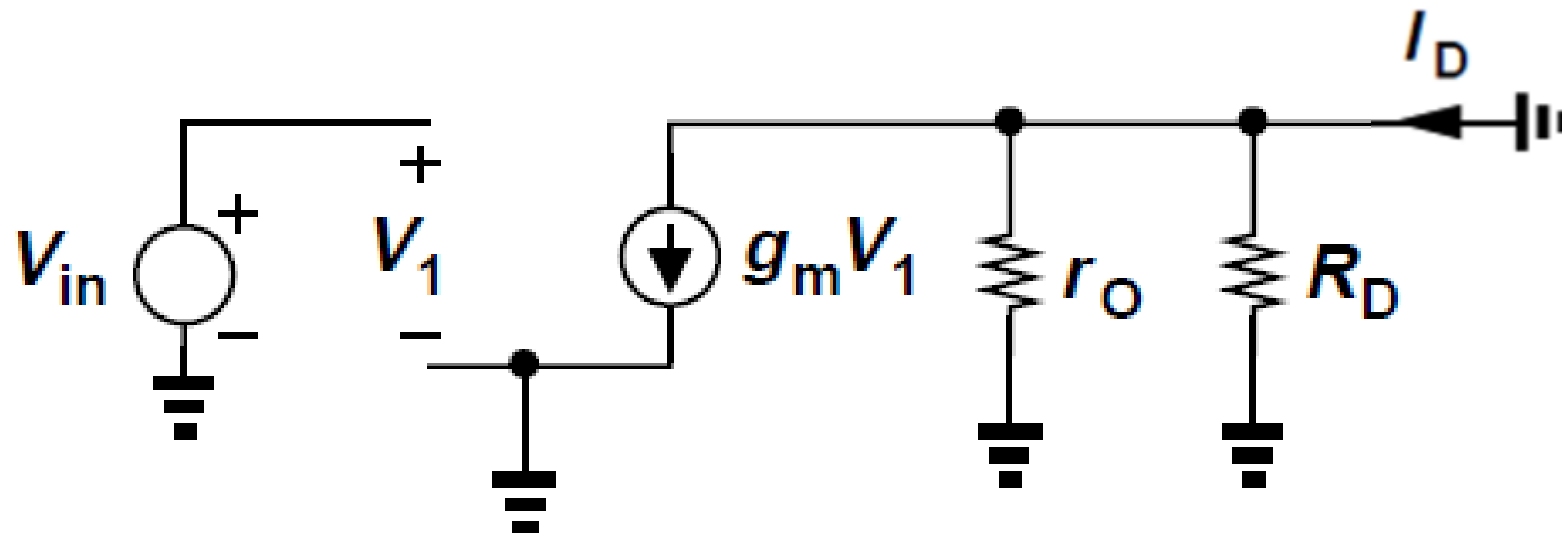
- To calculate the gain, we need to find g_m and r_{out} using small signal model.



- Note that for the circuit A, R_D is infinite and $10\text{ k}\Omega$ for the circuit B.

Theoretical Material: CS Stage Gain Calculation

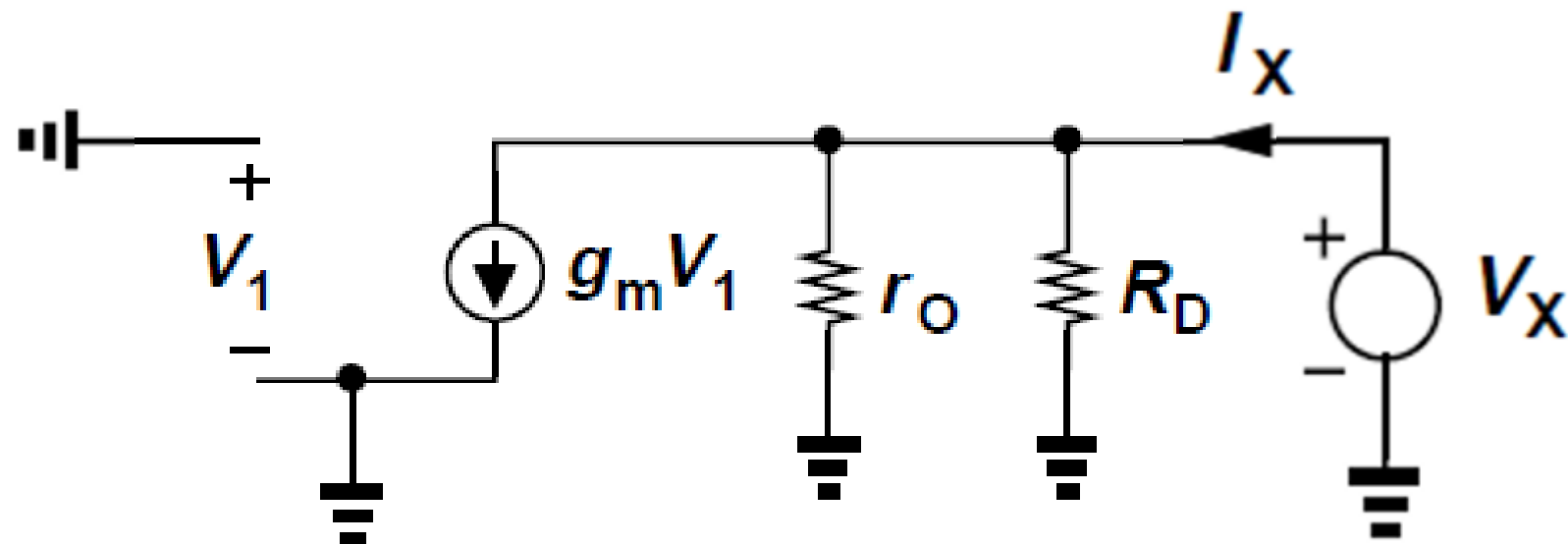
- To find g_m , we keep V_{in} and short circuit V_{out}



In this case, the transconductance of the circuit is simply g_m

Theoretical Material: CS Stage Gain Calculation

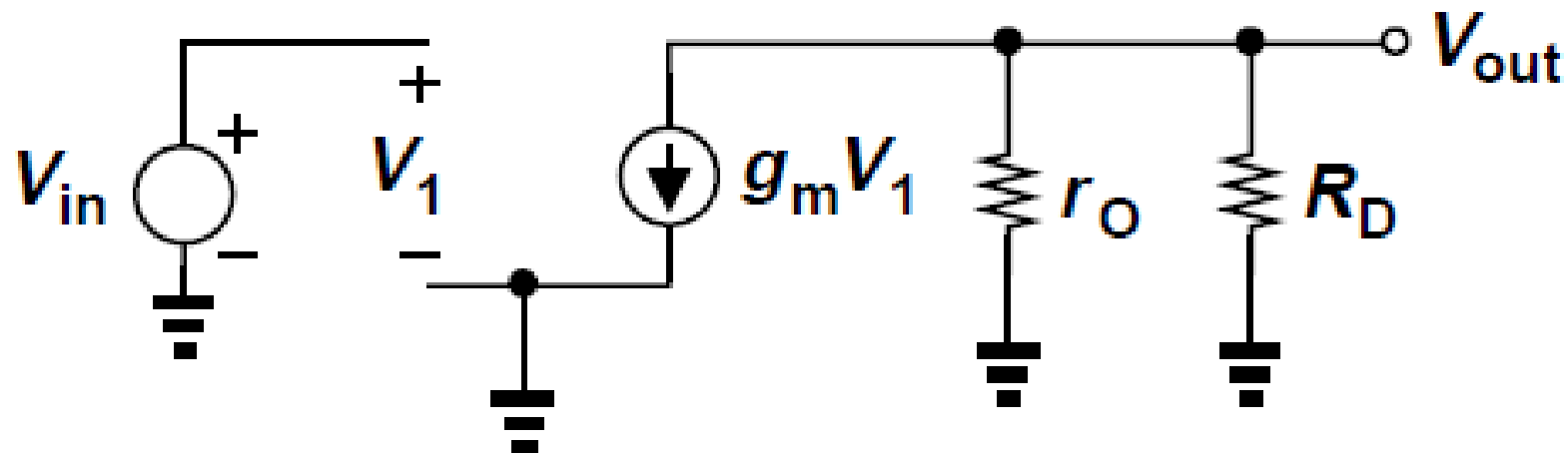
- To find R_{out} , we short V_{in} and apply V_x and measure I_x



$$R_{out} = \frac{r_o R_D}{r_o + R_D}$$

Theoretical Material: CS Stage Gain Calculation

- So the small signal voltage gain can now be found:



$$A_v = -g_m \frac{r_o R_D}{r_o + R_D}$$

Exercise 1a

From LTSpice error log, we can find all relevant parameters and calculate the absolute DC gain:

$$|A_{vA}| = g_{m1} * r_{o1} = 6.63e^{-5} * 1/1.39e^{-4} \approx 0.48 \text{ or } -6.4 \text{ dB}$$

$$\begin{aligned} |A_{vB}| &= g_{m2} * R_D || r_{o2} = 2.03e^{-4} * (R_D || 1/8.3e^{-7}) \\ &\approx 2.03e^{-4} * 10 \text{ k}\Omega = 2.03 \text{ or } 6.1 \text{ dB} \end{aligned}$$

Exercise 1a

Now let's run AC simulation:

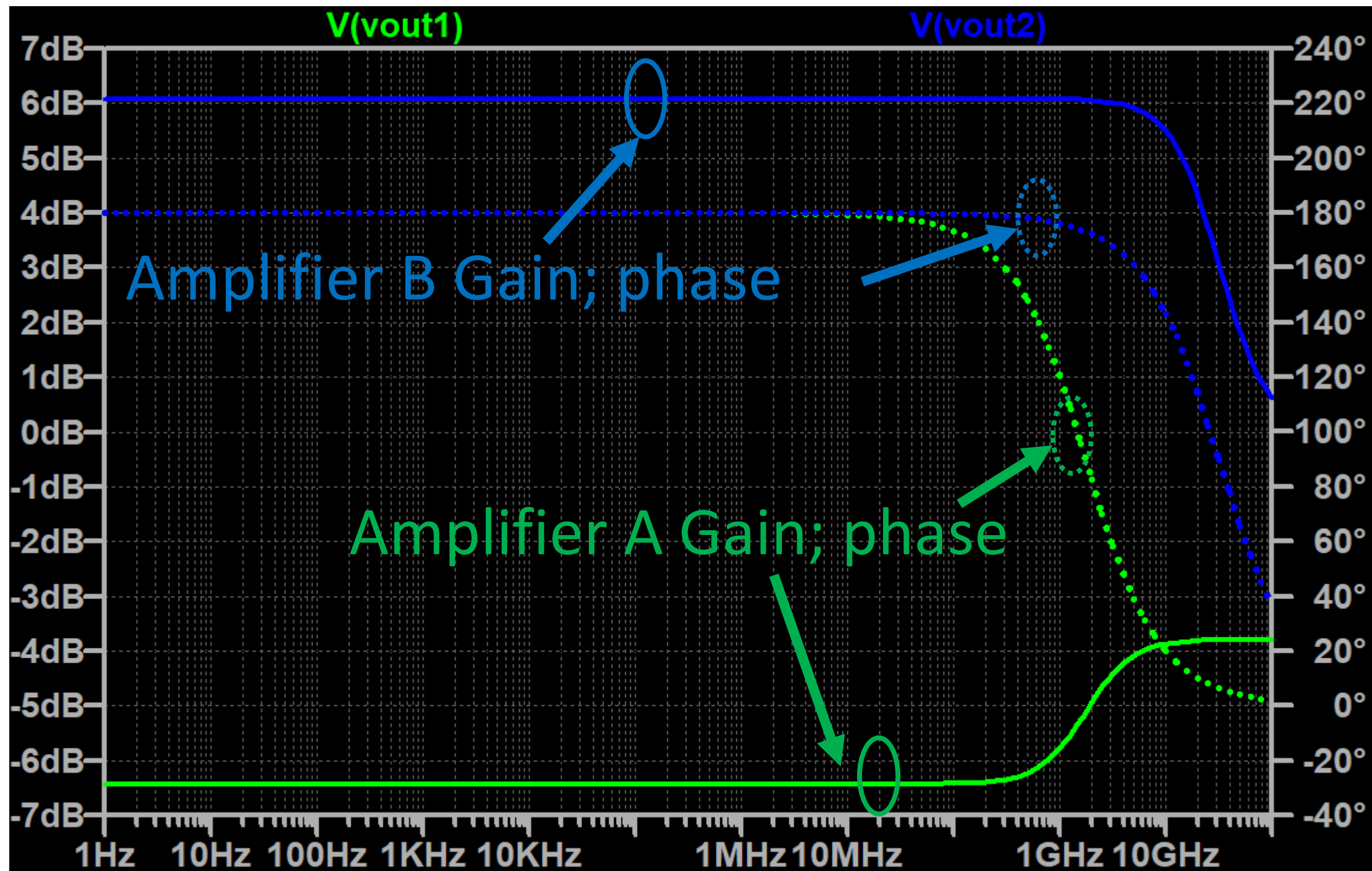
First edit the simulation text, use “;” in front of op to disable DC simulation and put “.” in front of AC simulation to enable AC simulation.

```
.lib 'C:\Program Files\LTC\LTspiceXVII\lib\cmp\log018.l' TT  
;op  
.ac dec 10 1 100G
```

Then click run!

And then left click on the output node to get the bode plot

Results



Bode plots of amplifier A and B

Exercise 1b

- Now change I_{D1} so that the drain current of M_1 equals to that of M_2
- Do a DC operating point simulation. Are both transistors in saturation?
- Calculate the DC gain of both amplifiers
- Do an AC simulation to plot the gain of the two amplifiers. Does the simulation agree with your calculation?

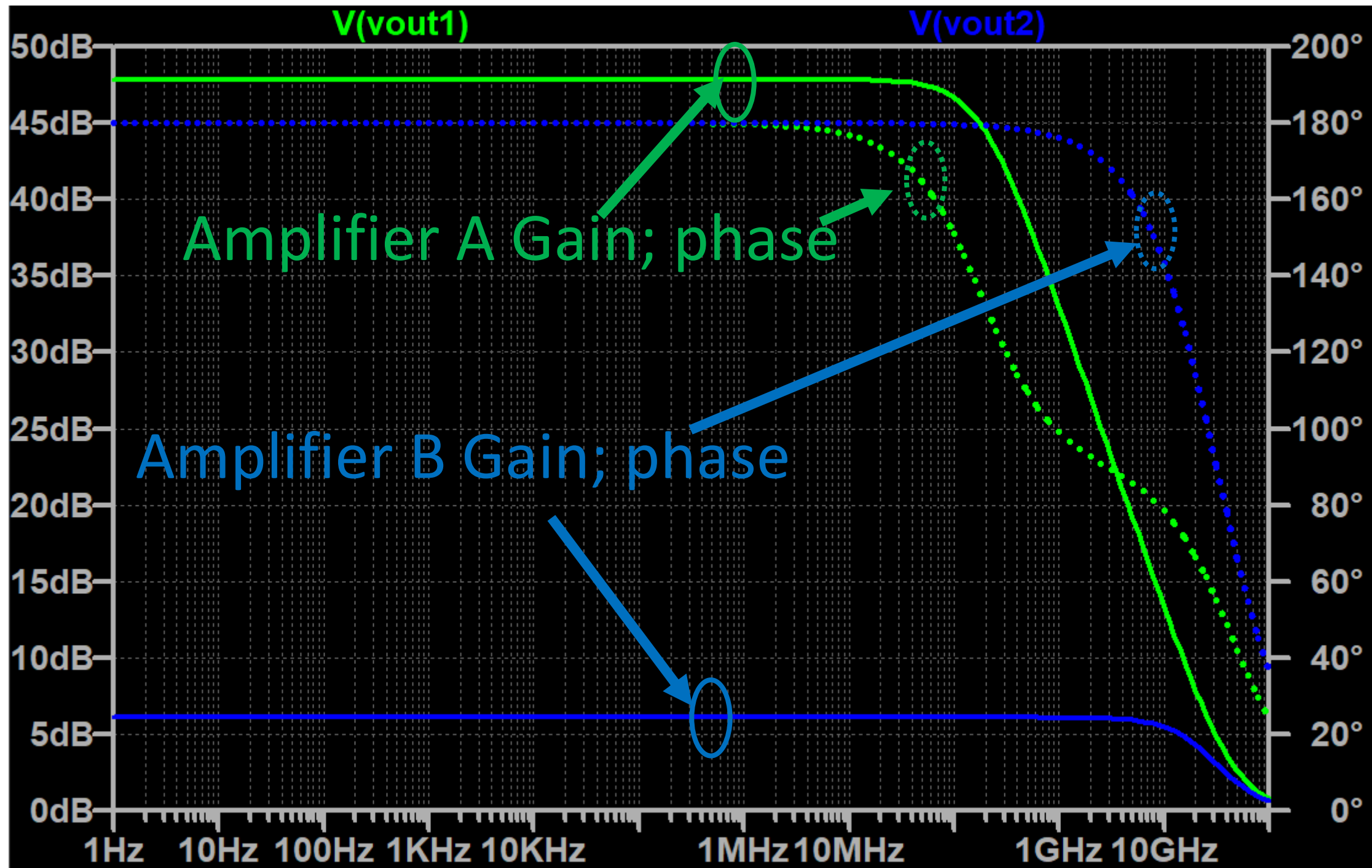
Exercise 1b

- From the previous exercise, we know I_{D2} is 38 μA . Now we change I_{D1} to 38 μA as well.
- After running a DC operating point simulation, we get:

$$|A_{vA}| = 2.03e^{-4} * (1/8.3e^{-7}) \approx 245 \text{ or } 48 \text{ dB}$$

- Note that now both amplifiers consume the same current, but the DC gain is quite different.

Results



Bode plot of amplifier A and B

Exercise 1 Summary

- This exercise shows that DC bias is crucial
- A CS stage with current source load can improve DC gain quite significantly

Exercise 2

- Open Exc2_v2.asc, you will see a resistive-degenerated CS stage with current source load. The current source is implemented by a PMOS transistor: M_{pc2} .
- Here we have a simple “current mirror: M_{pc1} and M_{pc2} . Since their gates are connected together, they have the same V_{gs} . And since they have also the same dimension, they have “**almost**” the same drain current. So we say M_{pc2} “mirrors” the current from M_{pc1} .
- Do a DC operating point simulation to find out whether all the transistors are in saturation.

Exercise 2

- First we run DC operating point simulation and open SPICE error log, and we see that all transistors are in saturation.

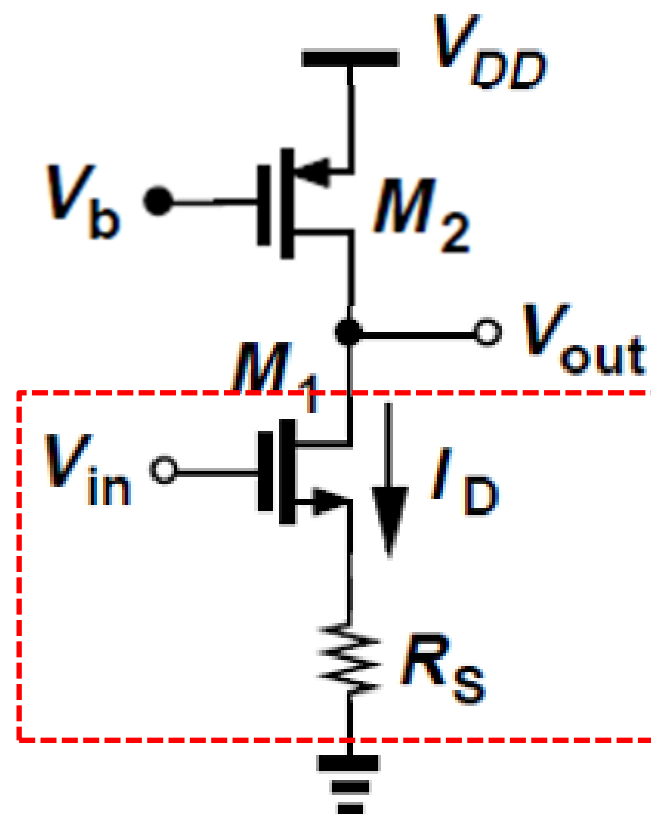
```
- --- BSIM3 MOSFETS ---  
Name:      m1      mpc2      mpc1  
Model:     nch.9    pch.8      pch.8  
Id:        1.00e-05 -1.00e-05 -1.00e-05  
Vgs:       6.30e-01 -1.10e+00 -1.10e+00  
Vds:       4.38e-01 -1.16e+00 -1.10e+00  
Vbs:      -2.00e-01  0.00e+00  0.00e+00  
Vth:       5.82e-01 -4.49e-01 -4.49e-01  
Vdsat:     9.07e-02 -5.32e-01 -5.32e-01  
Gm:        1.43e-04  2.69e-05  2.69e-05  
Gds:       4.63e-06  2.57e-07  2.74e-07  
Gmb        3.71e-05  9.40e-06  9.38e-06
```

Exercise 2

- Calculate the absolute DC gain of the amplifier
- Do an AC simulation to plot the gain and compare it with your calculation.

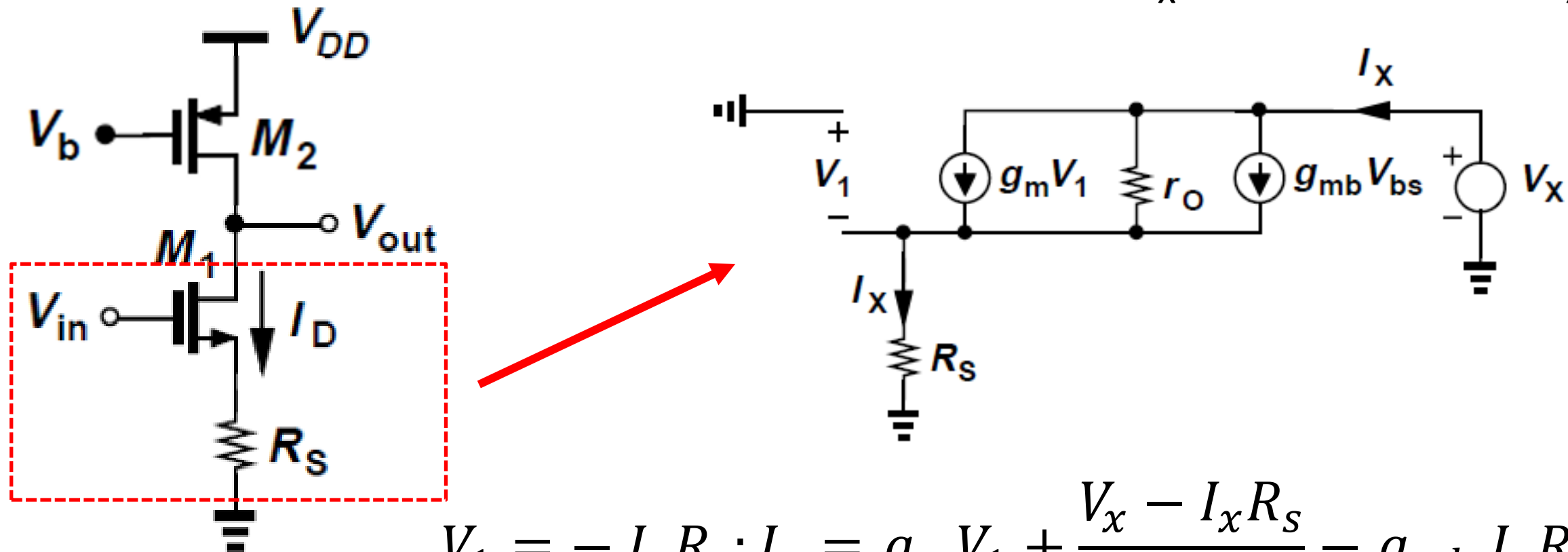
Exercise 2

- To calculate the DC gain of the circuit, we need to calculate its transconductance and output impedance.
- We already know M_2 only contributes to the output impedance of this circuit (r_{o2})
- Now we need to calculate the transconductance and output impedance of the circled part



Exercise 2

- Now let's use small signal model to calculate R_{out} of the circled part. We apply a voltage source V_x and measure I_x .



$$V_1 = -I_x R_S; I_x = g_m V_1 + \frac{V_x - I_x R_S}{r_o} - g_{mb} I_x R_S$$

$$R_{out} = \frac{V_x}{I_x} \Rightarrow R_{out} = [1 + (g_m + g_{mb})r_o]R_S + r_o$$

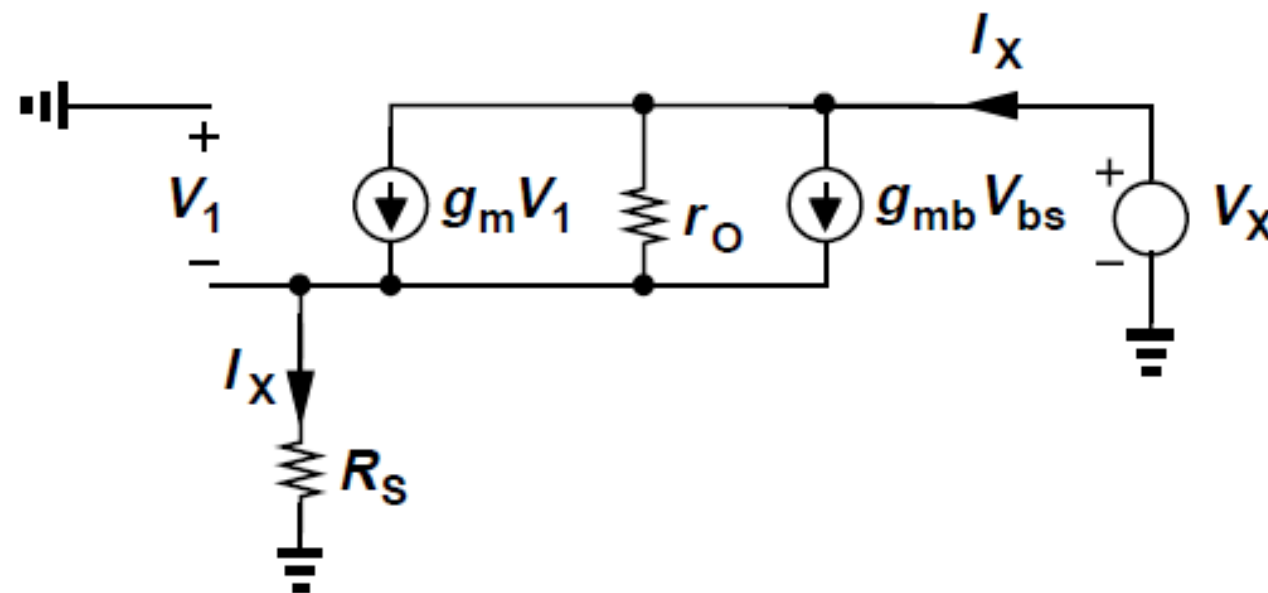
$$\text{If } (g_m + g_{mb})r_o \gg 1$$

$$R_{out} \approx (g_m + g_{mb})r_o R_S + r_o = A_{v1} R_S + r_o$$

Exercise 2

- From LTspice, we can find:
- $g_{m1} = 1.43e^{-4}$; $g_{mb1} = 3.71e^{-5}$; $r_{o1} = 1/4.63e^{-6}$; $R_s = 20 \text{ k}\Omega$
- So $(g_m + g_{mb})r_o \gg 1$, and we can use the following simplified equation:

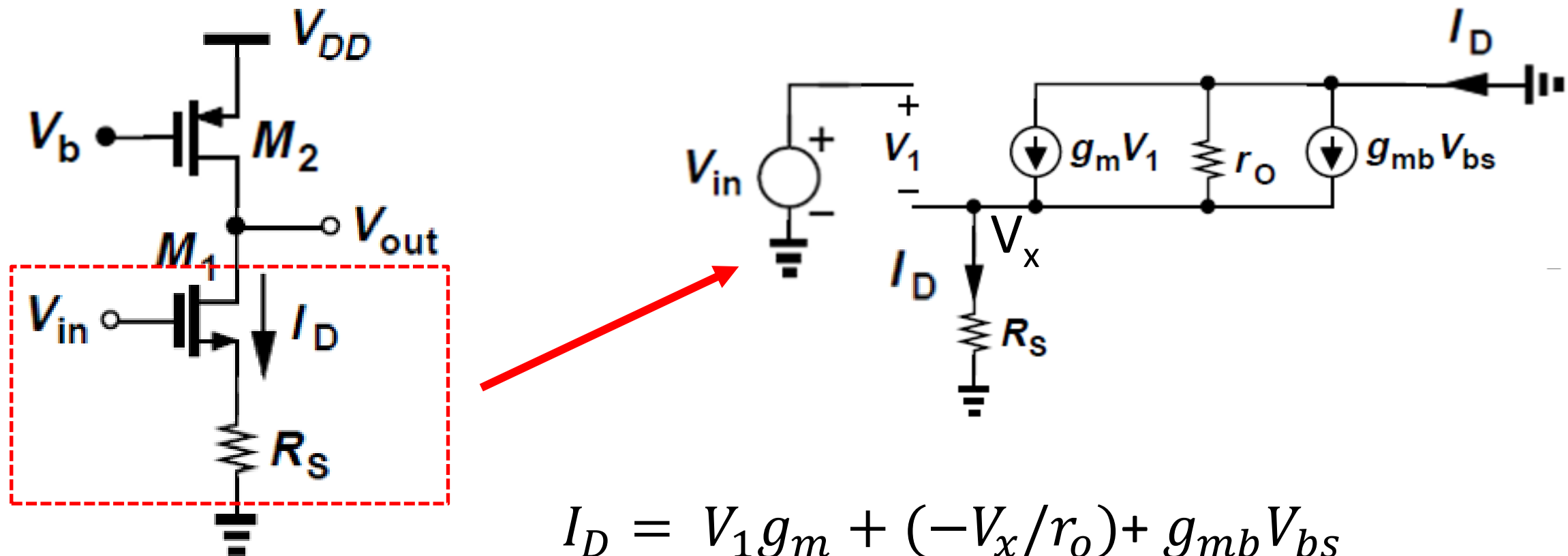
$$R_{out} \approx (g_m + g_{mb})r_o R_s + r_o = A_{v1} R_s + r_o$$



- Finally, we get $R_{out} \approx 1.0 \text{ M}\Omega$

Exercise 2

- Now let's calculate the equivalent G_m of the circled part. We apply an input signal and short circuit the output.



$$I_D = V_1 g_m + (-V_x / r_o) + g_{mb} V_{bs}$$

$$V_{bs} = -V_x = -I_D R_S ; V_1 = V_{in} - V_x$$

$$G_m = I_D / V_{in} = \frac{g_m}{1 + R_S g_m + \frac{R_S}{r_o} + g_{mb} R_S} \approx \frac{g_m}{1 + R_S g_m + g_{mb} R_S} \approx \frac{g_m}{1 + R_S g_m}$$

You can choose your approximation according to the real situation

Exercise 2

- From LTSpice error log, we get:

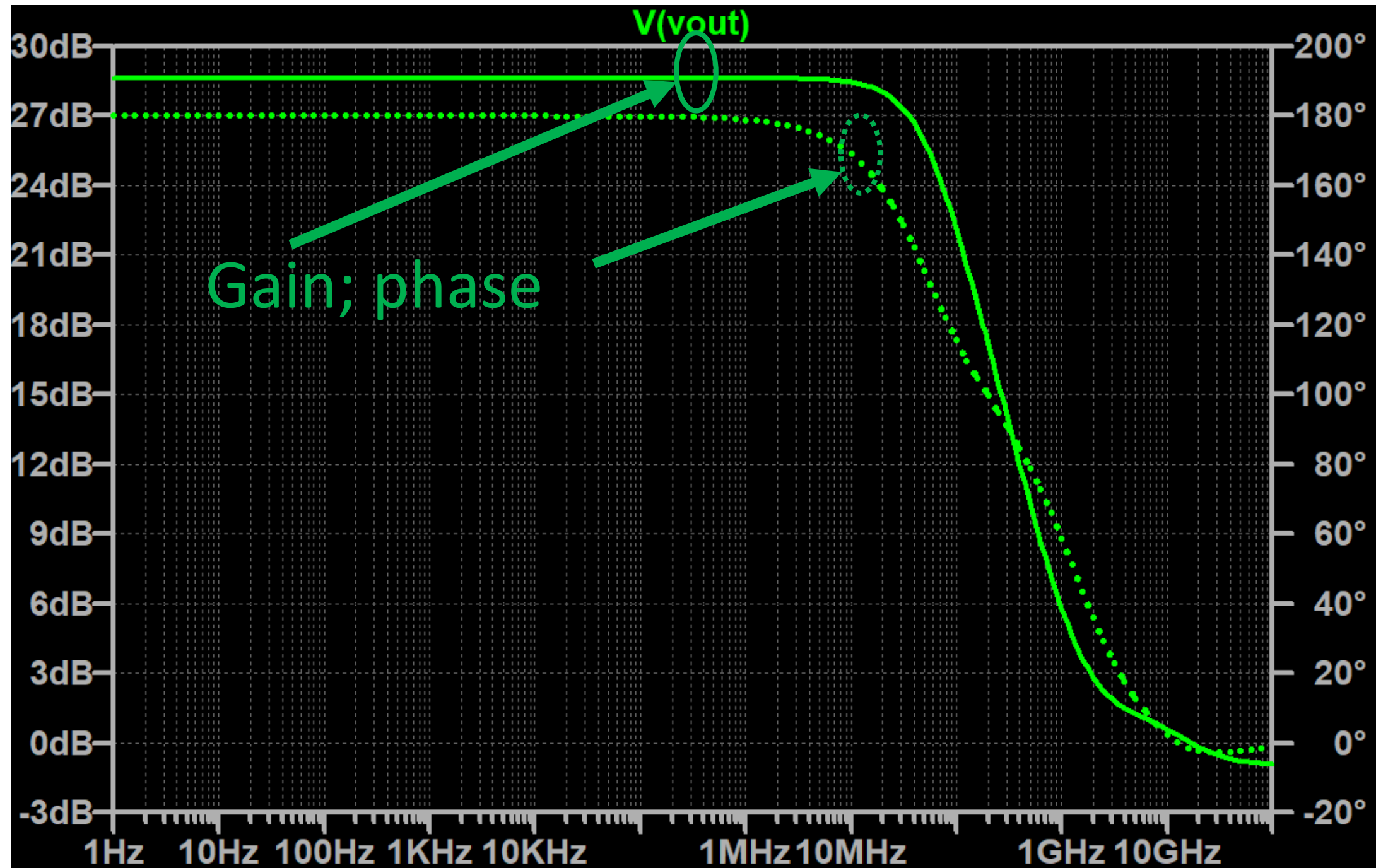
$$g_m(M_1) = 1.43e^{-4}; g_{mb}(M_1) = 3.71e^{-5}; R_s = 20 \text{ k}\Omega; r_{o1} = 1/4.63e^{-6}$$

- So we can say R_s/r_o is quite small and therefore is negligible. Let's take body effect into account,

$$G_m \approx \frac{g_m}{1 + R_s g_m + g_{mb} R_s} = \frac{1.43e^{-4}}{1 + 20 \text{ k}\Omega * 1.8e^{-4}} = 0.31e^{-4}$$

- With $R_{out} \approx 1.0 \text{ M}\Omega$ and $r_{o2} = 1/2.57e^{-7} \approx 3.9 \text{ M}\Omega$, the total output impedance of the amplifier is $R_{out} || r_{o2} = 0.8 \text{ M}\Omega$.
- Finally, $|A_v| = G_m * R_{outtotal} = -0.31e^{-4} * 0.8 \text{ M}\Omega \approx 24$ or 27.8 dB

Results



Bode plot of the resistive-degenerated amplifier